Database Technologies – Assignment 8 **KD2-Abhishek-96835**

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD2_Abhishek_96835>select * from orders where cnum in(select cnum from customers where cname='Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD2_Abhishek_96835>select o.amt,c.cname,c.rating from orders as o inner join customers as c on c.cnum=o.cnum where o.amt>(select AVG(amt) from orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.01 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD2_Abhishek_96835>select snum,sum(amt) total from orders group by snum having sum(amt)>(select MAX(amt) from orders);
+------+----------+
| snum | total    |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD2_Abhishek_96835>select cnum,cname from customers c where c.rating>= ANY(select rating from customers c inner join salespeople as s on c.snum=s.snum where
 s.sname='Serres');
+------+----------+
| cnum | cname    |
+------+----------+
| 2002 | Giovanni |
| 2003 | Liu      |
| 2004 | Grass    |
| 2008 | Cisneros |
+------+----------+
4 rows in set (0.00 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD2_Abhishek_96835>select * from salespeople s where s.city <> ALL (select c.city from customers c where s.city=c.city AND s.snum=c.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.01 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD2_Abhishek_96835>select * from orders where amt > ANY(select amt from orders where cnum in(select cnum from customers where city='London'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
KD2_Abhishek_96835>select * from orders o where o.snum=(select s.snum from salespeople s where s.sname='Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.01 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD2_Abhishek_96835>select * from orders o where snum in(select snum from customers c where c.city='London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD2_Abhishek_96835>select sname,snum from salespeople where snum in (select snum from customers group by snum having COUNT(cnum)>1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD2_Abhishek_96835>select s.sname,s.snum,s.city,(select COUNT(*) from customers c where c.snum=s.snum) as count from salespeople s where s.snum in (select s
num from customers group by snum having COUNT(*)>1);
+--------+------+----------+-------+
| sname  | snum | city     | count |
+--------+------+----------+-------+
| Peel   | 1001 | London   |     2 |
| Serres | 1002 | San Jose |     2 |
+--------+------+----------+-------+
2 rows in set (0.00 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD2_Abhishek_96835>select * from customers where rating > ANY(select rating from customers where city='Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
KD2_Abhishek_96835>select * from orders where amt > ANY(select MIN(amt) from orders where odate='1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD2_Abhishek_96835>select * from orders where amt < ANY(select amt from orders where cnum in(select cnum from customers where city='San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
KD2_Abhishek_96835>select * from customers where rating > ALL(select rating from customers where city='London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```